

Gap dynamics and disease-causing invasive species drive the development of an old-growth forest over 250 years

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ABSTRACT

Investigating the disturbance regimes of unharvested forests helps us understand their past, present, and future trajectory and gives us a model for forest management. It can also clarify the relative importance of small-scale gap dynamics *versus* more severe disturbances. Here we used tree rings to examine the recruitment patterns, growth dynamics, and disturbance chronologies of American beech (*Fagus grandifolia*), sugar maple (*Acer saccharum*), and eastern hemlock (*Tsuga canadensis*) in A.B. Williams Woods, an old-growth forest in Ohio, USA, over the past 250 years. We found that beech and sugar maple recruitment peaked around 1900 and continued through the 1900s, while hemlock recruitment peaked during 1825–1875, then declined and effectively ended in the early 1900s. Hemlock grew fastest during the 1800s according to ring width and basal area increment, while sugar maple ring width surpassed beech and hemlock in the 1900s. All three species showed a dramatic increase in growth from 1980 to 2010. Beech and sugar maple established regardless of canopy gaps, but 73% of hemlocks originated in gaps. In most decades, <10% of trees experienced gap recruitment or growth release, suggesting that ongoing, endogenous canopy mortality was the primary disturbance shaping this forest. However, a more severe forest-wide disturbance occurred during the 1980s–1990s when the scale insect causing beech bark disease was introduced, with greater than 30% of living trees showing growth releases in those decades. Another synchronous release occurred in the 1930s when blight-killed chestnuts were removed; 16% of trees showed releases. Both of these intermediate-severity disturbances involved human introduction of invasive species. Thus, we documented a natural disturbance regime of small-scale gap dynamics, punctuated by more severe anthropogenic disturbances in the twentieth century. These relatively frequent, intermediate-severity events probably mean that the forest's current composition is non-equilibrium. Hemlock may continue to decline while beech maintains its dominance by constant regeneration in both gaps and shade, and by responding to disturbance with root suckering and growth pulses. The codominance of sugar maple may be relatively recent, and perhaps temporary, as we found little sugar maple recruitment before 1875 or after 1950, and three times fewer sugar maple saplings now than in the early 1900s. Despite being protected as a park, the development of this old-growth forest has been shaped more by disease-causing invasive species than natural disturbances over the past century. This result emphasizes the pervasiveness of human impacts even in communities we look to as examples of natural pattern and process.

1. Introduction

Disturbance can transform the structure, composition, and diversity of ecological communities (Watt, 1947; Pickett and White, 1985; Frelich, 2002). Therefore, to understand how current communities have come to be and how they may change in the future, we need to examine the impacts of past disturbances. Moreover, management plans to sustain biodiversity in the long term aim to mimic natural disturbance regimes, making it imperative that we assess the frequency and intensity of

disturbances that shaped communities historically (Swetnam et al., 1999; Palik et al., 2002; Seymour et al., 2002; Lindenmayer et al., 2006). In the temperate forest biome, old-growth forests provide an ideal opportunity to study how plant communities have developed in response to natural disturbances and in the absence of intentional human manipulation (Whitney, 1987; McCarthy, 1995). Numerous studies have used the information contained in tree rings to ascertain the disturbance history of temperate deciduous or mixed old-growth forests (e.g. Frelich and Lorimer, 1991; Fraver et al., 2009; Nagel et al., 2014; Rubio-

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Cuadrado et al., 2018). The prevailing conceptual model based on this work suggests that forest development is driven primarily by individual tree mortality and small-scale gap dynamics (Bormann and Likens, 1979; Runkle, 1982, 2000). In particular, the beech-maple forests of eastern North America are thought to have an extremely low frequency of stand-replacing disturbances, primarily windstorms, with a return interval of 1000–3000 years (Lorimer and White, 2003). However, the relative importance of ongoing gap dynamics versus periodic, intermediate-severity disturbances like ice and windstorms and pest and disease outbreaks remains unclear.

Here we use tree rings to investigate the disturbance history of A.B. Williams Woods, an old-growth beech-maple-hemlock forest in north-east Ohio, USA, to understand how the forest has reached its current composition and predict how it will change in the future. In part, we hope to explain the long-term compositional changes we discovered via a resurvey study (Flinn et al., 2019). We found that the forest had undergone surprisingly substantial shifts from 1932 to 2018: American beech (*Fagus grandifolia* Ehrh.; hereafter, beech) and sugar maple (*Acer saccharum* Marshall; hereafter, sugar maple) appear to be displacing eastern hemlock (*Tsuga canadensis* (L.) Carrière; hereafter, hemlock). Since 1932, larger beech trees died and were replaced by numerous root suckers, increasing the overall stem density of the forest by 50%. Hemlock declined in frequency and stem density and produced virtually no saplings. Plots dominated by hemlock in 1932 shifted toward dominance by sugar maple, which increased greatly in frequency, stem density, and basal area. Reasons for these shifts remain unknown; examining the forest's disturbance history should provide insight into the mechanisms and causes of these compositional changes. At the same time, it addresses the larger question of the role of disturbance in the long-term stability and variability of unharvested forests.

In this study, we glean three types of information from tree rings. First, we examine the age structure of the three dominant species by estimating each tree's pith date. Second, we use ring widths and basal area increments to compare the growth of the three species over time. Third, we identify bursts of growth that indicate gap recruitment and release events, allowing us to reconstruct the frequency and severity of past disturbances. Our objective is to describe and compare the recruitment patterns, growth dynamics, and disturbance chronologies of beech, sugar maple, and hemlock in A.B. Williams Woods over the past 250 years. We discuss how interspecific differences in these patterns can explain the compositional changes ongoing in the forest, and how the long-term disturbance regime of this old-growth forest compares with widely accepted models dominated by small-scale gap dynamics.

2. Materials and methods

2.1. Study area

A.B. Williams Woods is a 26-ha old-growth forest located in Cleveland Metroparks's North Chagrin Reservation, Mayfield, Ohio, USA (41°33'40" N 81°25'39" W, 274 m elevation; Fig. 1). The region has a humid continental climate with a mean annual temperature of 10.8 °C, mean annual precipitation of 99.4 cm, including 173 cm of snow, and a median frost-free season of 189 days (Midwestern Regional Climate Center, 2017). The mostly level land lies on the glaciated Allegheny Plateau, with Cleveland shale underlying somewhat poorly drained clay loam soils (Williams 1936). No logging has occurred for at least the last 150 years except salvage logging of American chestnut killed by blight (Williams, 1936). In 2018, the stem density of trees ≥ 5 cm dbh was 795 stems/ha and basal area was 55 m²/ha (Flinn et al., 2019). The three

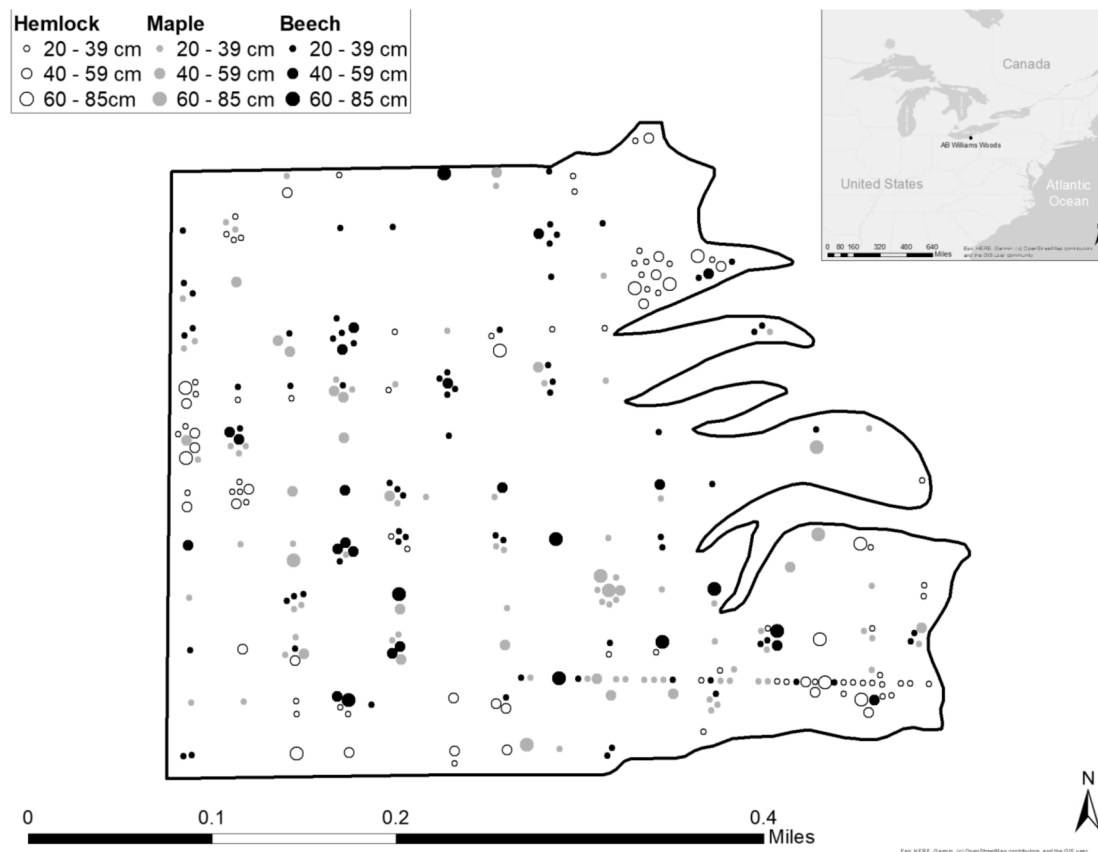


Fig. 1. Map of A.B. Williams Woods, Mayfield, Ohio, USA, showing its location in North America (inset), and the sizes (dbh in cm) and locations of the cored trees. We sampled the transect and plots of Williams (1936) as well as a regular grid. Points were moved apart slightly to facilitate visibility. The east side of the site borders the Chagrin River valley.

most abundant tree species were beech, representing 43% of all trees in the forest, sugar maple with 29%, and hemlock with 12% (Flinn et al., 2019).

2.2. Field sampling

To make our sample consistent with previous ecological research at the site over the past 89 years, we first collected tree cores in the 58 plots established by A.B. Williams in 1932 and resurveyed by us in 2018 (layout described in Williams, 1936; Flinn et al., 2019; Fig. 1). We centered 50-m² circular plots on these locations (the largest plot size that would prevent overlapping in adjacent transect plots). In each plot, we measured diameter at breast height (dbh; 1.37 m) and cored all living beech, sugar maple, and hemlock trees of ≥ 20 cm dbh. We used a 5.15 mm-diameter Hagl f increment borer and took two cores from each tree at breast height. When we encountered rot, we did not take a second core. In the same way, we then sampled 147 50-m² circular plots spread throughout the study area on a regular grid, spaced 44 m apart (Fig. 1). To increase sample sizes, we expanded a subset of 41 of these plots to 500 m². This yielded 187 cores of beech from 101 trees, 193 cores of sugar maple from 100 trees, and 195 cores of hemlock from 101 trees.

2.3. Dendroecological methods

We air-dried the cores, glued them on wooden mounts (Custom Woodwork by Billy Jack, Perkinston, Mississippi, USA), and sanded them by hand with progressively finer sandpaper until rings were clearly visible (Speer, 2010). To make digital images, we scanned the cores with at least 1200 dpi resolution. We measured ring widths to the nearest 0.01 mm using CooRecorder and CDendro software (Cybis Elektronik & Data AB, Saltsj baden, Sweden, <https://www.cybis.se/>). Rings were cross-dated visually using marker years and by comparison to a master chronology from a drought-sensitive chestnut oak forest 19 km from our study site (Lanier et al. 2018). For cores that did not hit the pith, we estimated the number of rings to pith using the curvature and mean width of the last five rings, only if the estimated number of rings to pith was ten or less. We counted or estimated pith dates for 71 beech, 62 sugar maple, and 42 hemlock trees. We used COFECHA to check and correct the cross-dating (Holmes, 1983; Grissino-Mayer, 2001). We omitted cores whose dating was uncertain due to problems like missing pieces. This yielded 180 confidently dated cores of beech from 101 trees, 189 cores of sugar maple from 99 trees, and 190 cores of hemlock from 99 trees. For beech, the series intercorrelation was 0.26, mean length 113 years, and total length 255 years; for sugar maple, the series intercorrelation was 0.28, mean length 99 years, and total length 209 years; for hemlock, the series intercorrelation was 0.55, mean length 129 years, and total length 236 years.

2.4. Data analysis

We used data from Flinn et al. (2019), where we measured trees ≥ 5 cm dbh, to assess size distributions, and we used trees ≥ 20 cm dbh with pith dates or estimates to assess age distributions. We considered recruitment age to be the year the tree reached breast height. To compare growth rates among species, we calculated median ring width and the median of the maximum and minimum 10-year rolling mean ring width for each core (Orwig and Abrams, 1994; Nagel et al., 2014). These ring widths incorporate the different size and age distributions of the species because ring width declines with increasing tree size and age (Visser, 1995; Biondi and Qeadan, 2008). In addition, because we sampled trees ≥ 20 cm dbh and used all of their ring widths, data from trees < 20 cm dbh is present throughout the dataset except in the most recent years. This could affect the pattern of ring widths over time if the growth of trees < 20 cm dbh differed consistently from trees ≥ 20 cm dbh. Therefore, to see how growth varied independent of size and age distributions, we also used basal area increment (BAI). BAI represents

the yearly net increase in the cross-sectional area of the tree stem, measured from the bark to the pith; it is calculated as $BAI = \pi (R_t^2 - R_{t-1}^2)$ where R is the stem radius at the end and beginning of an annual ring formed in years t and $t - 1$ (Biondi and Qeadan, 2008). Our calculations were based on diameter inside the bark, which we obtained using published relationships between dbh outside bark and dbh inside bark for each species (Hilt et al., 1983; Miles and Smith, 2009). These equations were prepared for mature but not old-growth trees, so they may overestimate diameter inside bark for very old trees. We compared dbh, age, and growth parameters using Kruskal-Wallis tests.

To construct disturbance chronologies for individual trees, we identified gap recruitment (i.e. rapid early growth rates) and release events (i.e. abrupt increases in growth), both of which indicate mortality of a former canopy tree (Lorimer and Frelich, 1989). To distinguish gap-recruited from understory-recruited trees, we compared the mean radial growth for years 1–20 with the mean radial growth for years 21–40 (Lorimer et al., 1988; McCarthy and Bailey, 1996; Hart et al., 2012; Goins et al., 2013; Rubio-Cuadrado et al., 2018). If the mean for the first 20 years exceeded the mean for the second 20 years, then we classified the tree as originating in a canopy gap. We compared the frequency of gap-recruited *versus* understory-recruited trees to random expectations using G tests, and we used a G test to compare the frequency of gap *versus* understory recruitment in the 1800s and 1900s.

To identify release events, we used a boundary line technique with radial growth averaging criteria (Nowacki and Abrams, 1997; Black and Abrams, 2003; Splechtna et al. 2005). In this approach, the mean radial growth over a prior 10-year period M_1 (including the target year) was compared to the mean radial growth over a subsequent 10-year period M_2 (excluding the target year) for all possible time intervals in each core. The percentage growth change over each interval, assigned to the target year, was computed as $\%GC = [(M_2 - M_1)/M_1] \times 100$ (Nowacki and Abrams, 1997). The maximum value of each growth pulse (positive, sequential percentage growth change) was selected as a potential release. To minimize detection of short-term growth responses caused by factors other than loss of canopy trees, such as climate extremes or mast years, we selected potential releases only if growth pulses had positive percentage growth change exceeding 50% (Splechtna et al. 2005; Svoboda et al. 2012). To further avoid detecting short-term events, we excluded releases that lasted < 10 years (Fraver and White, 2005; Drobyshev et al., 2010). We then constructed our own boundary line representing the maximum percentage growth change that is physiologically possible at a given level of prior growth—thus accounting for effects of tree age, diameter, and canopy position on the magnitude of release response—and described each percentage growth change value as a fraction of its maximum potential, given its level of prior growth (Black and Abrams, 2003). We identified events falling within 20–100% of the boundary line as releases (Black and Abrams, 2003).

We binned release events by decade to allow for the variable lag period between disturbance and tree response (Lorimer and Frelich, 1989; Fraver et al., 2009). We built species-specific and forest-wide disturbance chronologies by assessing the percentage of trees affected by new gap formation (gap recruitment or release) in each decade (Frelich and Graumlich, 1994). This type of population-based disturbance chronology represents the severity of disturbance rather than the canopy area disturbed (Frelich, 2002; Busby et al., 2008;  ada et al., 2016). We truncated chronologies when the number of cores dropped below ten (Fraver et al., 2009). Our method did not detect releases in 2010–2019 because it required a 10-year post-disturbance window. Gap recruitment events in recent decades may also have been underestimated because trees had not yet reached 20 cm dbh. To distinguish exogenous disturbances like windstorms, ice storms, insect outbreaks, and pathogen attacks, which cause synchronous gaps, from ongoing, endogenous gap-phase dynamics, we differentiated species-specific or forest-wide gap recruitment and release events from individual-level events. We identified decades when $\geq 25\%$ of trees of one species

experienced gap recruitment or release as “species-specific” disturbances, and decades when $\geq 25\%$ of all trees experienced gap recruitment or release as “forest-wide” disturbances (Lorimer, 1980; Nowacki and Abrams, 1997; Rubino and McCarthy, 2004). We felt justified in calling these events “forest-wide” because beech, sugar maple, and hemlock comprised 84% of trees in the forest (Flinn et al., 2019). We calculated the percentages of release events that were individual, species-specific, and forest-wide. To make a rough estimate of disturbance return intervals, we divided the number of decades with a given level of disturbance by the total number of decades; the numerical inverse is the return interval (Fraver et al., 2009). We conducted analyses in Excel (Microsoft, Redmond, Washington, USA) and R (R Core Team, 2021), using the packages zoo (Zeileis and Grothendieck, 2005), dplR (Bunn, 2008), and TRADER (Altman et al., 2014).

3. Results

3.1. Size distributions

The size distributions of beech and sugar maple showed the negative exponential or reverse-J pattern typical of uneven-aged forests (Fig. 2). The majority of these trees, 62% of beech and 51% of sugar maples, belonged to the smallest size class with dbh < 10 cm. By contrast, hemlock had a more normal-shaped distribution, lacking in the smallest size classes, with a peak in the 15–30 cm dbh range. The median size of beech trees was slightly smaller than sugar maple, while the median size of hemlock trees was over twice as large as both beech and sugar maple (Table 1).

3.2. Age distributions

Age distributions of the dominant tree species confirmed the forest was uneven-aged (Fig. 3). Despite the forest's centuries-long history, most of the trees ≥ 20 cm dbh recruited during the century 1825–1925. The oldest individual tree we sampled was a beech with a pith date of 1756 (263 years old). Beech recruitment peaked in 1875–1900 and continued throughout the twentieth century, including the youngest sampled tree, recruited in 1985 (we cored trees ≥ 20 cm dbh). The oldest sampled sugar maple recruited in 1801. Sugar maple recruitment increased in the years 1875–1900 and showed a sharp peak in 1900–1925. Sugar maple continued to have substantial recruitment through 1950, with the youngest sampled sugar maple dated to 1968. The oldest hemlock, recruited in 1775, had a dbh of only 40 cm. Hemlock recruitment occurred most during 1825–1875, then declined and effectively ended in the early twentieth century. The youngest hemlock in our sample recruited in 1928. Overall, hemlocks were oldest, with a median recruitment year of 1856; beeches had a median recruitment year of 1888; and sugar maples were younger than beech trees but not significantly, with a median recruitment year of 1910 (Table 1).

3.3. Growth rates

Beech and sugar maple had greater ring widths and a higher 10-year maximum ring width for each core than hemlock (Table 1). Sugar maple had higher 10-year minimum ring widths, while beech and hemlock were similar. Patterns of BAI differed; sugar maple and hemlock had greater BAIs than beech (Table 1). The 10-year maximum BAI was similar among species, while sugar maple and hemlock had higher 10-year minimum BAIs than beech.

Up to circa 1870, hemlocks had a higher mean ring width than beech or sugar maple (Fig. 4a). From 1900 to 1980, sugar maple ring width surpassed beech and hemlock. All three species, but particularly beech, increased dramatically in ring width during 1980–2010. Considering BAI, hemlocks again had the greatest mean growth until circa 1900, with a distinct peak near 1850 (Fig. 4b). The BAI of beech and sugar maple increased throughout the time frame. Again, according to BAI, all three

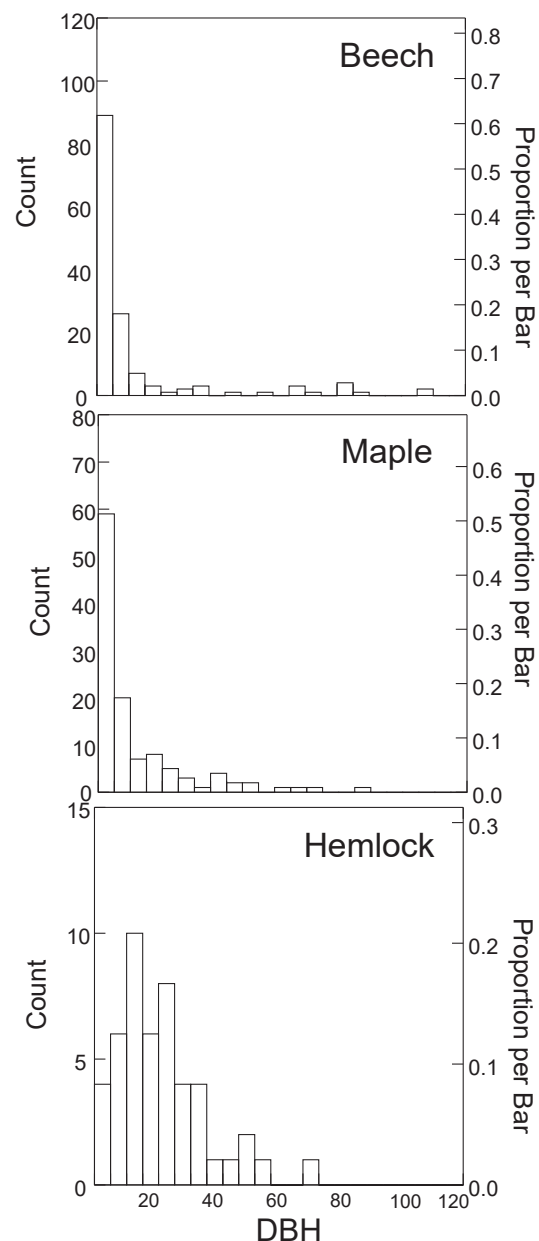


Fig. 2. Size distributions (diameter at breast height, 1.37 m, in cm) of the three dominant tree species in A.B. Williams Woods, Mayfield, Ohio, USA. Distributions show all trees ≥ 5 cm dbh sampled by Flinn et al. (2019), including 144 beech (*Fagus grandifolia*), 115 sugar maple (*Acer saccharum*), and 48 hemlock (*Tsuga canadensis*). The Y axes differ.

species showed a dramatic increase in growth from 1980 to 2010.

3.4. Disturbance history

While 63% of beech trees and 37% of sugar maples originated in canopy gaps, G tests indicated that these patterns did not differ from random; beech and sugar maple established regardless of canopy gaps. Hemlocks, however, established more often in canopy gaps than in the understory, with 73% of hemlocks originating in gaps ($G = 4.71$, $df = 1$, $P = 0.0300$). Most gap recruitment of beech and hemlock occurred in the mid-1800s (Fig. 5). During the 1800s, 63% of all recruitment occurred in canopy gaps, whereas in the 1900s, less than half of recruitment events were in gaps ($G = 3.53$, $df = 1$, $P = 0.0603$).

In most decades (79%), <10% of trees experienced gap recruitment

Table 1

Radial growth patterns of beech (*Fagus grandifolia*), sugar maple (*Acer saccharum*), and hemlock (*Tsuga canadensis*), the three dominant tree species in A.B. Williams Woods, Mayfield, Ohio, USA.

Parameter	Beech	Maple	Hemlock
Median dbh (cm)	8.7a (1.7)	9.9b (1.5)	24.6c (2.0)
Median age (years)	132a (5.1)	110a (3.3)	164b (5.5)
Ring width (mm/yr)			
Median	0.91a (0.007)	1.11a (0.007)	0.76b (0.008)
Median 10-yr maximum	2.68a (0.062)	2.65a (0.067)	2.07b (0.090)
Median 10-yr minimum	0.25a (0.034)	0.41b (0.018)	0.20a (0.016)
BAI (mm ² /yr)			
Median	334a (6.8)	491b (6.9)	489c (5.1)
Median 10-yr maximum	1630a (87)	1678a (94)	1368a (88)
Median 10-yr minimum	39a (28)	97b (16)	96b (15)

Notes: Medians in a row with different letters are significantly different according to pairwise comparisons after Kruskal-Wallis tests. Standard errors are shown in parentheses. Dbh data are from Flinn et al. (2019) where we sampled all trees ≥ 5 cm dbh, including 144 beech, 115 sugar maple, and 48 hemlock. Age data are from trees ≥ 20 cm dbh for which cores hit or nearly hit the pith, including 71 beech, 62 sugar maple, and 42 hemlock. Growth rate data are from all cores including 180 beech, 189 sugar maple, and 190 hemlock. “Median 10-year maximum” is the median, across all cores of a species, of the maximum 10-year rolling mean for each core. “Median 10-year minimum” is the median, across all cores of a species, of the minimum 10-year rolling mean for each core.

or release (Fig. 5). We identified one severe forest-wide disturbance during the 1980s–1990s. More than half of beech trees experienced release in the 1980s and 37% in the 1990s, along with 45% of sugar maples in the 1990s and 29% of hemlocks in the 1980s. There was also a marked, synchronous response by 16% of all trees in the 1930s. This event affected hemlocks most, with 22% having releases. Most gap recruitment and releases (70%) occurred synchronously in the 1930s and 1980s–1990s; none were species-specific, and 30% were individual. Excluding the synchronous events of the 1930s and 1980s–1990s, the mean percentage of trees experiencing gap recruitment or release was 6% per decade. The return interval for disturbances of $\geq 10\%$ of trees was 47 years; for disturbances of $\geq 25\%$ of trees, 70 years.

4. Discussion

4.1. Frequency, intensity, and agents of disturbance

The disturbance history of A.B. Williams Woods, along with its uneven age structure and dominance by shade-tolerant, late-successional species, suggests a disturbance regime driven by small-scale gap dynamics. This type of disturbance regime may characterize north-temperate beech-dominated forests in general (Splechna et al., 2005; Hobi et al., 2015). Most decades had a baseline rate of 6% of trees affected by disturbance, which makes sense given estimates from other old-growth forests of 5.7–6.9% (Lorimer and Frelich, 1989), 6.2% (Runkle, 1982), or 6.4% (Fraver et al., 2020) canopy area lost per decade. We found no evidence of stand-replacing disturbance over the past 250 years; no disturbance affected more than 35% of all trees. This supports our finding that severe, stand-level disturbances rarely occurred in the landscape before European settlement, as only 0.23% of lines surveyed circa 1800 had windfalls (Flinn et al., 2018). It also agrees with previous evidence that stand-replacing disturbances are very infrequent, on the order of every 1000–3000 years, in this forest type and throughout the hemlock-hardwood forests of the Great Lakes (Frelich and Lorimer, 1991; Lorimer and White, 2003). In addition, we found no evidence of fire over the past 250 years, including before the era of fire suppression. There were also no burned areas described on circa 1800 land surveys (Whitney, 1982; Flinn et al., 2018). This agrees with multiple other lines of evidence that many forests in the eastern United States, including the beech-maple type, were little affected by fire before European settlement (Matlack, 2013).

Despite the preeminence of small-scale gap dynamics in the development of the forest, we observed two intermediate-severity disturbances that caused significant mortality in the forest. The forest-wide release event in the 1980s–1990s followed the introduction of the scale insect that causes beech bark disease (*Cryptococcus fagisuga* Lind.). The insect was first detected in Ohio in 1984 in a forest 19 km from our study site (Mielke et al., 1985; MacKenzie and Iskra, 2005). A fungus typically associated with beech bark disease (one of two *Neonectria* species) was detected at our study site in 1993 (R. Tyler, personal communication). The result was numerous, large, standing dead beech trees surrounded by vast numbers of root sprouts (K.M. Flinn, personal observation). Because we only sampled living trees, we detected the effects of this

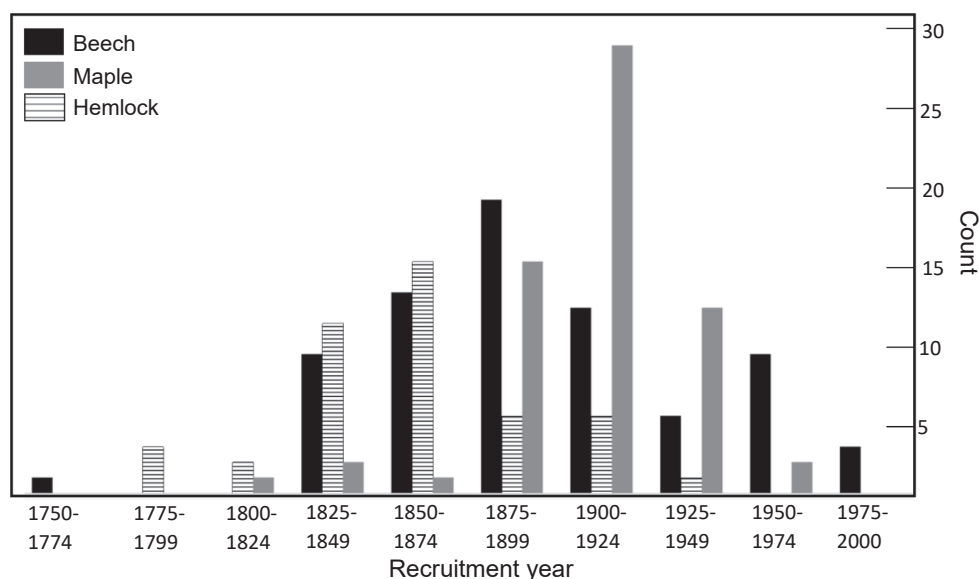


Fig. 3. Age distributions (recruitment age, at 1.37 m) of the three dominant tree species in A.B. Williams Woods, Mayfield, Ohio, USA. We cored trees ≥ 20 cm dbh. Distributions show trees for which cores hit or nearly hit the pith, including 71 beech (*Fagus grandifolia*), 62 sugar maple (*Acer saccharum*), and 42 hemlock (*Tsuga canadensis*).

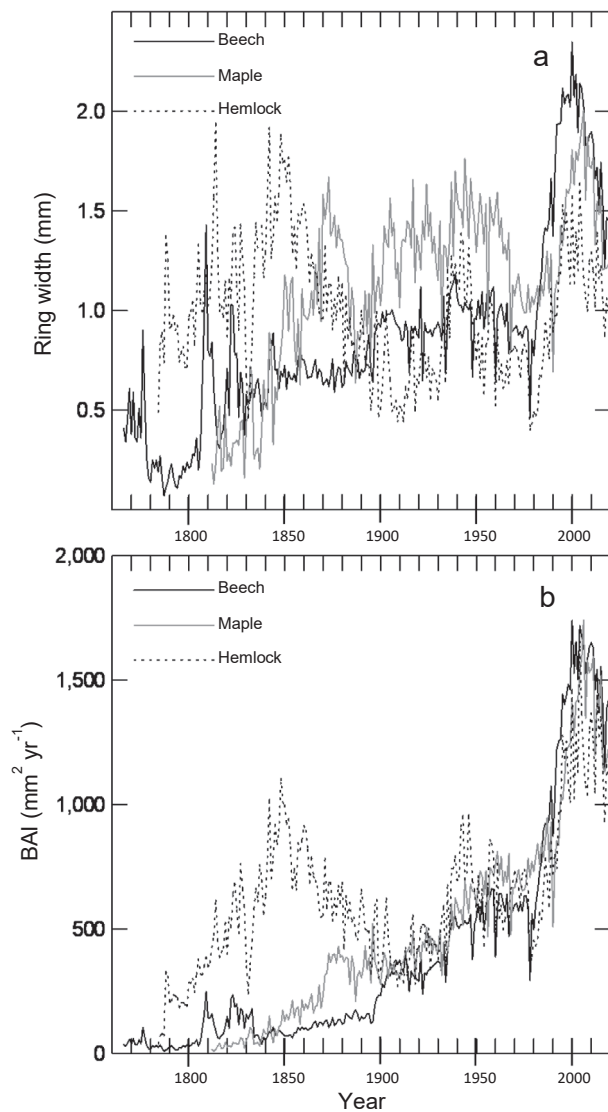


Fig. 4. Mean ring-width (a) and basal area increment (b) chronologies for the three dominant tree species in A.B. Williams Woods, Mayfield, Ohio, USA. Chronologies include 180 cores of beech (*Fagus grandifolia*), 189 of sugar maple (*Acer saccharum*), and 190 of hemlock (*Tsuga canadensis*).

mortality on surviving trees.

The other synchronous disturbance, in the 1930s, followed the removal of chestnut trees (*Castanea dentata* (Marshall) Borkh) killed by fungal blight (*Cryphonectria parasitica* (Murrill) Barr; Williams, 1936). Prior to the blight, chestnut comprised up to 15% of trees in some areas of the forest (Williams, 1936). It makes sense that hemlock would be most affected by their death and removal as, at the time, the areas of the forest with the most chestnut were dominated by hemlock and red oak (*Quercus rubra* L.; Williams, 1936). A.B. Williams's notes and photographs (Baldwin Wallace: CMNH A.B. Williams Collection, 2022) indicate that the salvage logging by the Civilian Conservation Corps involved a substantial amount of disturbance. Both of the intermediate-severity disturbances we documented involved human introduction of invasive species. The proximity of A.B. Williams Woods to urban Cleveland may make it especially prone to such impacts. However, host-specific insect outbreaks were the major disturbance in remote northern Maine as well (Fraver et al., 2009), and Cohen et al. (2016) found that above-background tree mortality due to insects, diseases, and physiological stress became the dominant disturbance in United States forests beginning in the mid-1990s. The forest's small size, its well-used trails,

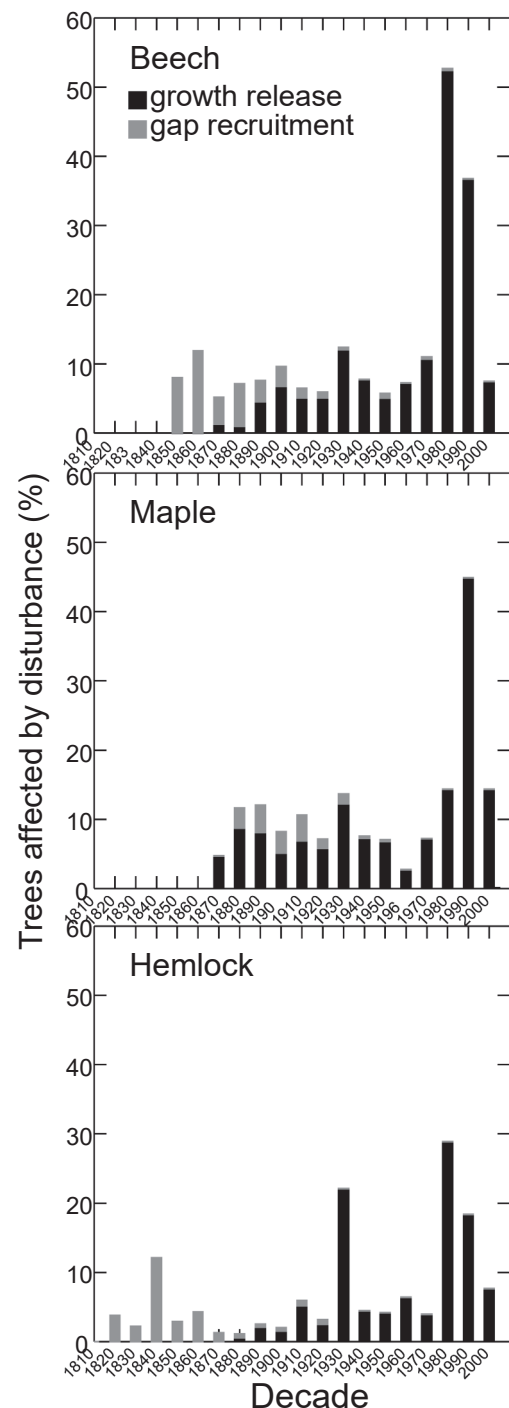


Fig. 5. Disturbance chronologies for the three dominant tree species in A.B. Williams Woods, Mayfield, Ohio, USA. Bars show the percentage of trees affected by disturbance (as evident in either gap recruitment or release) for each decade. Chronologies include 101 trees of beech (*Fagus grandifolia*), 99 of sugar maple (*Acer saccharum*), and 99 of hemlock (*Tsuga canadensis*).

and its surroundings, including roads, parking lots, and other developments of a public park, may also leave trees more vulnerable to disease and decline. Due to these events, intermediate-severity disturbances appeared to recur frequently over the past 250 years, on the order of every 50 years.

4.2. The forest's past, present, and future trajectory

Tree rings provide several lines of insight into how and why the forest is changing over time. Beech and sugar maple are displacing hemlock in this forest primarily because hemlock has had almost no regeneration since 1928. By 2018, there was only one hemlock sapling per hectare (<5 cm dbh; Flinn et al., 2019). Most of the existing hemlocks established in canopy gaps in the mid-1800s; the cause of the prevalence of canopy opening around 1850 is unknown, but young hemlocks capitalized on it with a pulse of establishment and rapid growth. Since then, beech and sugar maple recruited in the understory while hemlock largely did not. Hemlock has long been considered the most shade-tolerant of all North American tree species (Baker, 1949; Burns and Honkala, 1990), but in this forest, surprisingly, it has failed to regenerate in shade. The stem density of trees ≥ 5 cm dbh increased from 530 stems/ha in 1932 to 795 stems/ha in 2018 (Flinn et al., 2019), which likely further reduced light availability on the forest floor. This forest and the park system containing it have not yet been affected by hemlock woolly adelgid (*Adelges tsugae* Annand), although the invasive insect has recently been reported from our county (Ohio Department of Agriculture, 2021; C.E. Hausman, personal communication). Hemlock woolly adelgid could become an important pest in this forest in the near future. Hemlock's decline could also result from factors beyond the community level. In northeast Ohio, hemlock is on the western edge of its geographic range and occurs primarily in ravines; some see its local presence as a relict of post-glacial vegetation (Black and Mack, 1976). Though hemlock trees take up to 300 years to reach maturity and can live more than 800 years (Burns and Honkala, 1990), the species may continue to decline here. Due to hemlock's value to birds and other wildlife, and its uncommonness in northeast Ohio, this would represent a loss for both local and regional biodiversity (Tingley et al. 2002).

Beech has remained the most frequent, abundant, and important tree species in the forest throughout the twentieth and twenty-first centuries so far (where frequency is the percentage of plots where the species occurred, abundance is measured by stem density, and importance reflects basal area; Flinn et al., 2019). In this way, A.B. Williams Woods has remained representative of regional vegetation before European settlement, when beech dominated 47% of the landscape (Flinn et al., 2018). Here we see how beech has maintained its dominance—by constant regeneration in both gaps and shade, and by responding to beech bark disease with a huge pulse of root suckering and growth. Beech's unique sprouting ability allows it to regenerate successfully after multiple types of disturbance (Jones and Raynal, 1986; Morris et al., 2004; Busby et al., 2008). In effect, beech experienced a partial niche shift from an overstory to an understory tree after beech bark disease, similar to the niche shift of American chestnut after chestnut blight (Burke, 2012). Another factor that could favor beech regeneration over other species is that beech is less palatable to white-tailed deer than sugar maple or hemlock; regional deer populations declined during the 1800s to zero around 1900, but have since increased to densities far above those considered healthy for forest understories (C.E. Hausman, personal communication). Since 2014, the forest has also been heavily affected by the novel beech leaf disease associated with the nematode *Litylenchus crenatae mccannii*, first described from Japan (Carta et al., 2020; Martin and Volk, 2021; C.E. Hausman, personal communication). This disease is known to kill beech trees of all ages, but mortality is more common in smaller trees (Martin and Volk, 2021). The pattern of impacts on growth and mortality from this new beech disease may determine the future of the forest.

Sugar maple's codominance of the forest appears to be a relatively recent phenomenon; we found little sugar maple recruitment before 1875. Its pulse of recruitment around the turn of the twentieth century, along with its rapid growth, have led to sugar maple's current status as nearly equal to beech in frequency and second in abundance and basal area (Flinn et al., 2019). Interestingly, the rise of sugar maple within the forest coincides with similar increases in frequency and dominance

across northeast Ohio (Flinn et al., 2018) and throughout the northeast and upper midwest United States (Whitney, 1990; Schulte et al., 2007; Thompson et al., 2013). However, the abundance of sugar maple may not continue at the same level, as we found little recruitment after 1950, and there were three times fewer sugar maple saplings in 2018 than in 1932 (<5 cm dbh; Flinn et al., 2019).

Despite their contrasting histories of recruitment and growth, the dominant species in A.B. Williams Woods showed some common patterns as well. All three species demonstrated an overall increase in ring width and BAI over the time frame of the study, with a more pronounced increase after 1980. Though not well-documented in the literature, this overall increase has been observed in old-growth and second-growth chronologies throughout Ohio (Lanier et al., 2018; G. Wiles, personal communication). It may result from regional-scale factors like carbon dioxide enrichment or land-use changes leading to fertilization with windblown sediments (Ireland and Booth, 2012). In addition, though the major disturbances we observed were species-specific pathogens (chestnut blight and beech bark disease), the releases in response to these events were forest-wide and similar among species.

4.3. Conclusions

In this study, we documented a natural disturbance regime of ongoing, endogenous gap dynamics, punctuated by more severe anthropogenic disturbances in the twentieth century. These relatively frequent, intermediate-severity events probably mean that the forest's current composition is non-equilibrium. Models of community dynamics based on constant, low-intensity canopy mortality (Runkle, 1982; Woods, 1984) may be sufficient to explain the development of forests like A.B. Williams Woods over much of their history, but they cannot account for the recent anthropogenic disturbances these forests have experienced. Despite being protected as a park since 1925, the development of this old-growth forest has been shaped more by disease-causing invasive species than natural disturbances over the past century. This result emphasizes the pervasiveness of human impacts even in the communities we look to as examples of natural pattern and process.

CRediT authorship contribution statement

Kathryn M. Flinn: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Supervision, Funding acquisition. **Madison N. Dolnicsek:** Methodology, Investigation, Visualization, Writing – review & editing. **Abigail L. Cox:** Methodology, Investigation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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